SOLUTION

1.	Formula	Name	Comments
a.	CuCl	Copper(I) chloride	Because the anion is Cl^-, the cation must be Cu^+ (for charge balance), which requires a Roman numeral I
b.	HgO	Mercury(II) oxide	Because the anion is O^{2-}, the cation must be Hg^{2+} [mercury(II)]
c.	Fe_2O_3	Iron(III) oxide	The three O^{2-} ions carry a total charge of 6–, so two Fe^{3+} ions [iron(III)] are needed to give a 6+ charge

SOLUTION

1. Given the following systematic names, write the formula for each compound:
   a. Manganese(IV) oxide
   b. Lead(II) chloride

2.	Name	Formula	Comments
a.	Manganese(IV) oxide	MnO_2	Two O^{2-} ions (total charge 4–) are required by the Mn^{4+} ion [manganese(IV)]
b.	Lead(II) chloride	$PbCl_2$	Two Cl^- ions are required by the Pb^{2+} ion [lead(II)] for charge balance

INTERACTIVE EXAMPLE

1. NAME EACH OF THE FOLLOWING COMPOUNDS:
   a. PCL_5
   b. PCL_3
   c. SO_2

1.	Formula	Name
a.	PCl_5	Phosphorus pentachloride
b.	PCl_3	Phosphorus trichloride
c.	SO_2	Sulfur dioxide

2. FROM THE FOLLOWING SYSTEMATIC NAMES, WRITE THE FORMULA FOR EACH COMPOUND:
   a. SULFUR HEXAFLUORIDE
   b. SULFUR TRIOXIDE
   c. CARBON DIOXIDE

2.	Name	Formula
a.	Sulfur hexafluoride	SF_6
b.	Sulfur trioxide	SO_3
c.	Carbon dioxide	CO_2

INTERACTIVE EXAMPLE -NAMING COMPOUNDS CONTAINING POLYATOMIC IONS

1. Give the systematic name for each of the following compounds:

   a. N_2SO_4 c. $Fe(NO_3)_3$ e. Na_2SO_3

   b. KH_2PO_4 d. $Mn(OH)_2$ f. Na_2CO_3

2. Given the following systematic names, write the formula for each compound:

   a. Sodium hydrogen carbonate d. Sodium selenate

   b. Cesium perchlorate e. Potassium bromate

   c. Sodium hypochlorite

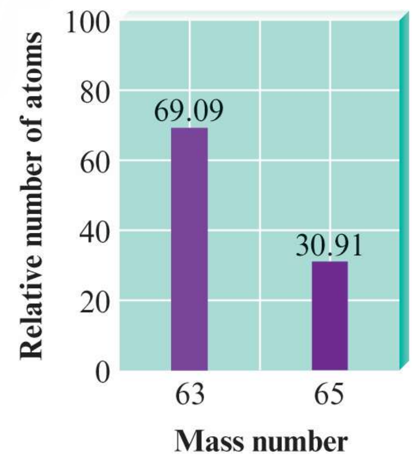
1.	Formula	Name	Comments
a.	Na_2SO_4	Sodium sulfate	
b.	KH_2PO_4	Potassium dihydrogen phosphate	
c.	$Fe_3(NO_3)_3$	Iron(III) nitrate	Transition metal—name must contain a Roman numeral. The Fe^{3+} ion balances three NO_{3-} ions.
d.	$Mn(OH)_2$	Manganese(II) hydroxide	Transition metal—name must contain a Roman numeral. The Mn^{2+} ion balances three OH^- ions
e.	Na_2SO_3	Sodium sulfite	
f.	Na_2CO_3	Sodium carbonate	

2.	Name	Formula	Comments
a.	Sodium hydrogen carbonate	$NaHCO_3$	Often called sodium bicarbonate
b.	Cesium perchlorate	$CsClO_4$	
c.	Sodium hypochlorite	NaOCl	
d.	Sodium selenate	Na_2SeO_4	Atoms in the same group, such as sulfur and selenium, often form similar ions that are named similarly. Thus, SeO_4^{2-} is selenate, like SO_4^{2-} is sulfate.

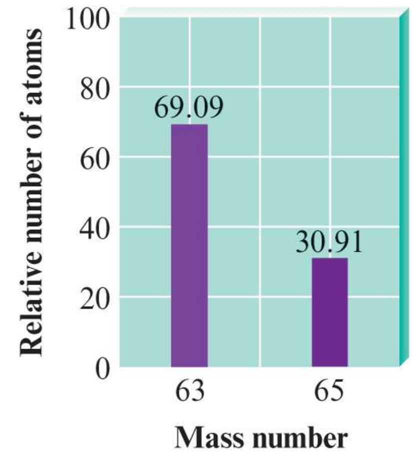
e) Potassium bromate: $KBrO_3$

EXAMPLE : AVERAGE MASS OF AN ELEMENT

- When a sample of natural copper is vaporized and injected into a mass spectrometer, the results shown in the graph are obtained
  - Use these data to compute the average mass of natural copper
    - Mass values for ^{63}Cu and ^{65}Cu are 62.93 u and 64.93 u, respectively



Solution



- Where are we going?
  - To calculate the average mass of natural copper
- What do we know?
  - ^{63}Cu mass = 62.93 u
  - ^{65}Cu mass = 64.93 u
- How do we get there?
  - As shown by the graph, of every 100 atoms of natural copper, 69.09 are ^{63}Cu and 30.91 are ^{65}Cu

Solution

- Thus, the mass of 100 atoms of natural copper is

$$\left(69.09\ \cancel{\text{atoms}}\right)\left(62.93\ \frac{\text{u}}{\cancel{\text{atom}}}\right) + \left(30.91\ \cancel{\text{atoms}}\right)\left(64.93\ \frac{\text{u}}{\cancel{\text{atom}}}\right) = 6355\ \text{u}$$

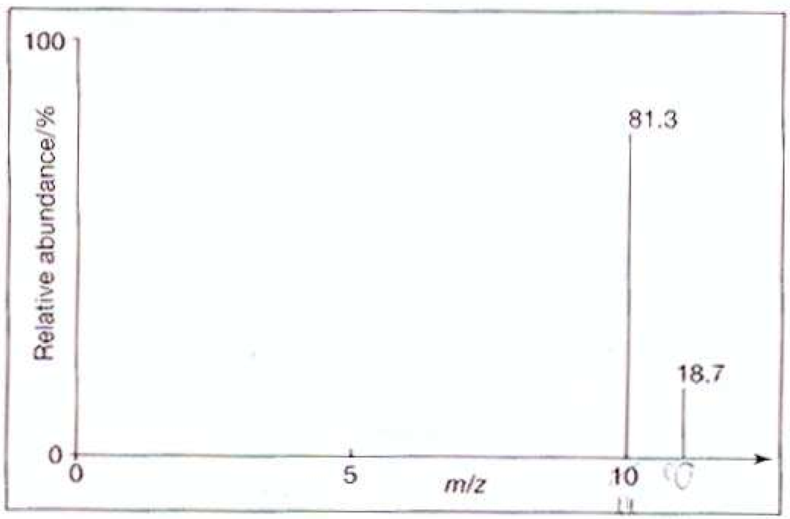
- Average mass of a copper atom is

$$\frac{6355\ \text{u}}{100\ \text{atoms}} = 63.55\ \text{u/atom}$$

  - This mass value is used in doing calculations involving the reactions of copper

Exp 1: Mass Spectrometer

- Isotopes of boron



m/z value	11	10
Relative abundance %	18.7	81.3

$$A_r \text{ of boron} = \frac{(11 \times 18.7) + (10 \times 81.3)}{(18.7 + 81.3)}$$

$$= \frac{205.7 + 813}{100}$$

$$= \frac{1018.7}{100} = \underline{10.2}$$

Exp 2: Mass Spectrometer

Bromine has two naturally occurring isotopes with the following percentage natural abundances:

Isotope	Natural abundance /%
^{79}Br	50.69
^{81}Br	49.31

Calculate the relative atomic mass of bromine correct to two decimal places.

The relative atomic mass is the weighted average of the atomic masses of the isotopes and their relative abundance. Therefore:

$$A_r = 79 \times \frac{50.69}{100} + 81 \times \frac{49.31}{100} = 79.9862 \approx 79.99$$

Exp 3: Mass Spectrometer

- Natural carbon is a mixture of ^{12}C, ^{13}C, and ^{14}C
  - Atomic mass of carbon is an average value of these three isotopes
- Composition of natural carbon:
  - ^{12}C atoms (mass = 12 u) - 98.89%
  - ^{13}C atoms (mass = 13.003355 u) - 1.11%

- Calculation of average atomic mass for natural carbon

  98.89% of 12 u + 1.11% of 13.0034 u

  $= (0.9889)(12\text{ u}) + (0.0111)(13.0034\text{ u})$

  $= 12.01\text{ u}$

  - For stoichiometric purposes, assume that carbon is composed of only one type of atom with a mass of 12.01

Exp 4: Mass Spectrometer

MASS SPECTROMETER – **QUESTIONS**

- A MASS SPEC CHART FOR A SAMPLE OF NEON SHOWS THAT IT CONTAINS:
  - 90.9% ^{20}NE
  - 0.17% ^{21}NE
  - 8.93% ^{22}NE

  CALCULATE THE RELATIVE ATOMIC MASS OF NEON

  YOU MUST SHOW ALL YOUR WORKING!

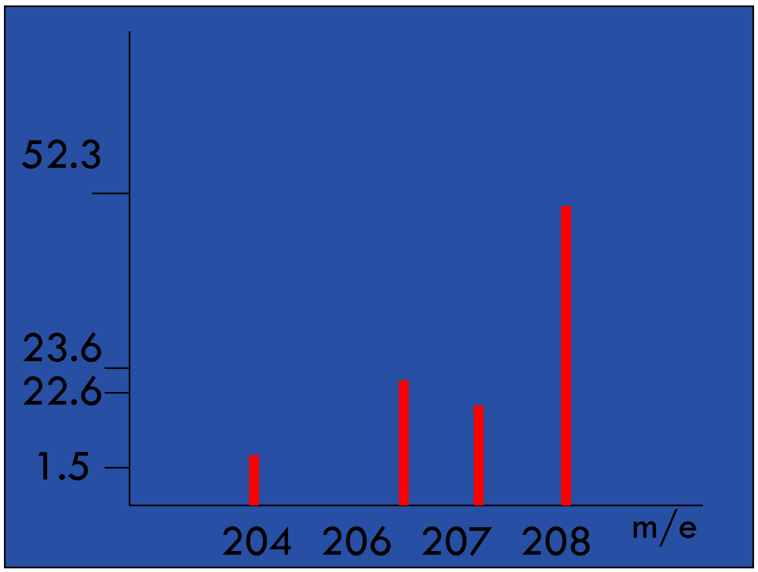
MASS SPECTROMETER – **QUESTIONS**

- 90.9% ^{20}NE
- 0.17% ^{21}NE
- 8.93% ^{22}NE

$$\frac{(90.9 \times 20) + (0.17 \times 21) + (8.93 \times 22)}{100}$$

$A_r = 20.18$

Exp 5: Mass Spectrometer



CALCULATE THE RELATIVE ATOMIC MASS OF LEAD

YOU MUST SHOW ALL YOUR WORKING!

MASS SPECTROMETER – **QUESTIONS**

- 1.5% ^{204}PB
- 23.6% ^{206}PB
- 22.6% ^{207}PB
- 52.3% ^{208}PB

$$\frac{(1.5 \times 204) + (23.6 \times 206) + (22.6 \times 207)+(52.3 \times 208)}{100}$$

$$\frac{306 + 4861.6 + 4678.2 + 10878.4}{100} \qquad \frac{20724.2}{100}$$

$A_r = 207.24$

Exp 6: Mass Spectrometer

EXAMPLE: FIND THE RELATIVE ATOMIC MASS (A_R) OF NATURALLY OCCURRING LEAD FROM THE DATA BELOW. RECORD YOUR ANSWER TO THE NEAREST TENTH.

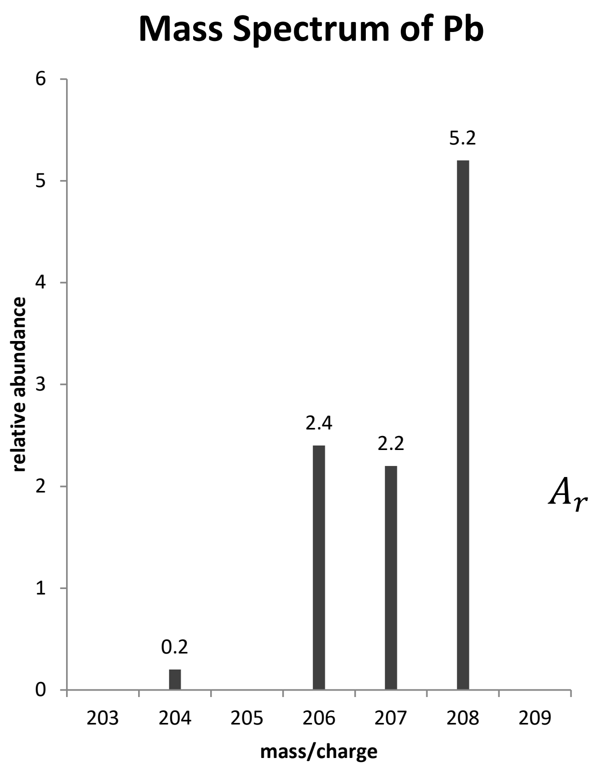


Figure: The Mass Spectrum of Naturally Occurring Lead

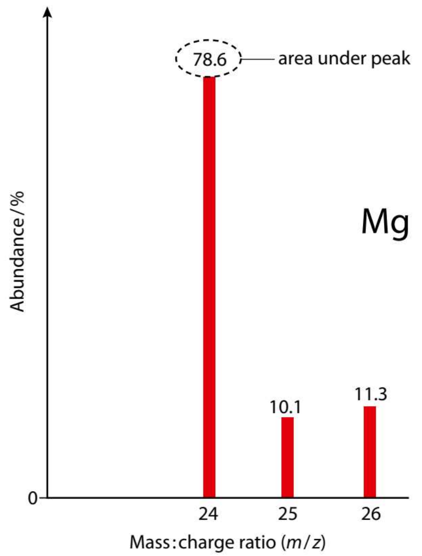
Isotopic mass	Relative abundance	% relative abundance
204	0.2	2
206	2.4	24
207	2.2	22
208	5.2	52

$$A_r = \frac{(2 \times 204) + (24 \times 206) + (22 \times 207) + (52 \times 208)}{100}$$

$$A_r = 207.2$$

Exp 7: Mass Spectrometer

Mass Spectrum of Magnesium



- The mass spectrum of Mg shows that Mg consists of 3 isotopes: ^{24}Mg, ^{25}Mg and ^{26}Mg.
- The height of each line is proportional to the abundance of each isotope.
- ^{24}Mg is the most abundant of the 3 isotopes.

The relative atomic mass can be calculated using:

$$A_r = \frac{(78.6 \times 24) + (10.1 \times 25) + (11.3 \times 26)}{100} = 24.3$$

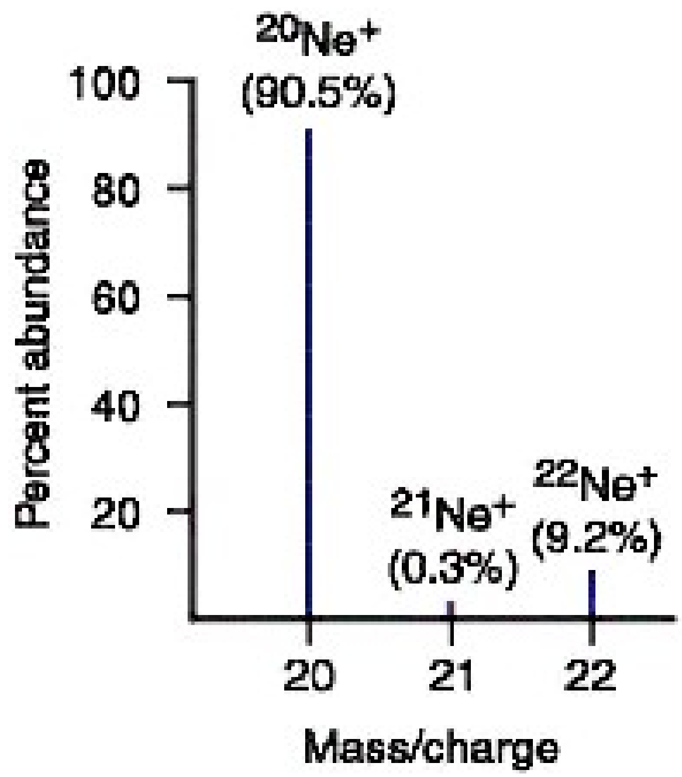
How to calculate the relative atomic mass from mass spectrum?

- **RAM** is calculated using data from the **mass spectrum.**

$$Average\ atomic\ mass = \frac{\sum(Isotopic\ mass\ x\ Abundance)}{\sum Abundance}$$

Question 1:

Calculate the relative atomic mass of neon from the mass spectrum.



Relative atomic mass of Ne

$$= \frac{(20\ x\ 90.5) + (21\ x\ 0.3) + (22\ x\ 9.2)}{(90.5 + 0.3 + 9.2)}$$

$$= \frac{2018.7}{100}$$

$$= 20.2 \text{ amu}$$

Question 2:

Copper occurs naturally as mixture of 69.09% of ^{63}Cu and 30.91% of ^{65}Cu. The isotopic masses of ^{63}Cu and ^{65}Cu are 62.93 *amu* and 64.93 *amu* respectively. Calculate the relative atomic mass of copper.

Relative atomic mass of Cu

$$= \frac{(62.93\ x\ 69.09) + (64.93\ x\ 30.91)}{(69.09 + 30.91)}$$

$$= \frac{6354.82}{100}$$

$$= 63.55 \text{ amu}$$

Question 3:

Naturally occurring iridium, Ir is composed of two isotopes, ^{191}Ir and ^{193}Ir in the ratio of 5:8. The relative isotopic mass of ^{191}Ir and ^{193}Ir are 191.021 *amu* and 193.025 *amu* respectively. Calculate the relative atomic mass of Iridium.

Relative atomic mass of Ir

$$= \frac{(191.021\ x\ 5) + (193.025\ x\ 8)}{(5+8)}$$

$$= \frac{2499.325}{13}$$

$$= 192.256 \text{ amu}$$